



Difference between Software Engineer, Developer, Programmer and other confusing terms

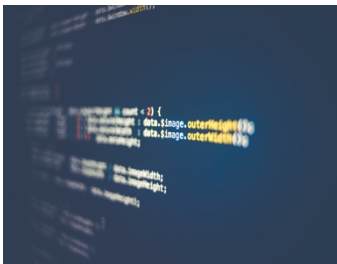
Software Engineer



A software engineer is a person **who** applies the principles of software engineering to the design, development, maintenance, testing, and evaluation of the software that make computers or other devices containing software work.

As you can see, the previous definition covers a broad range of duties, which goes from development or programming to configurations, testing, maintenance and other concerns related to software. In **this way**, a Software Engineer is a more general term you can find, and by extension, it applies to the other definitions we'll cover next.

Software Developer



A software developer is a person concerned with facets of the software development process, including the research, design, programming, and testing of computer software.

You can see a Software Developer as an engineer who specializes in the first 4 phases of the Software Development Life Cycle (For a complete and formal definition, take a look at IEEE's Software Engineering Body of Knowledge SWEBOK)

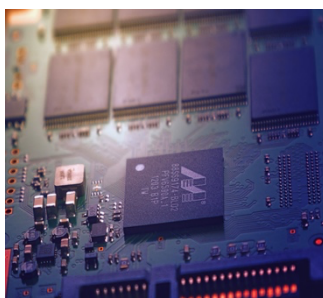
Computing Engineer

This one mainly is a source of confusion to many people (including me once upon a time). Perhaps the word computing make many of us think **of it** as a synonym of software or just the use of a computer to get work done. Some others even think that's about repairing computers, which is, in fact, a professional duty rather than engineering.

To clear our minds, let's read the following definition,

Computer engineering is a discipline that integrates several fields of electrical engineering and computer science required to develop computer hardware and software.[1] Computer engineers usually have training in electronic engineering (or electrical engineering), software design, and hardware-software integration instead of only software engineering or electronic engineering. Computer engineers are involved in many hardware and software aspects of computing, from the design of individual microcontrollers, microprocessors, personal computers, and supercomputers, to circuit design.

As you can see, it's the cross point for two different but still related disciplines, Software, and Electronics Engineering. (Recently I wrote a post about moving from Electronics to Software Engineering you can read here)



Electronics Engineer

Electronics engineering, or electronic engineering, is an electrical engineering discipline which utilizes nonlinear and active electrical components (such as semiconductor devices, especially transistors, diodes and integrated circuits) to design electronic circuits, devices, microprocessors, microcontrollers and other systems. The discipline typically also designs passive electrical components, usually based on printed circuit boards.

Electronics engineers usually have programming skills in low-level programming languages, **which** lets **them** become easily in Software Engineers, for instance in Embedded Software Development.



1. Lea el texto. Enumera las líneas.

2. Claves de la lectura Comprensiva. Escriba 2 ejemplos de cada uno:

- a. Indicaciones tipográficas: _____
- b. Cogandos. _____
- c. Palabras repetidas _____

3. a. Indica si las siguientes oraciones son Verdaderas o Falsa. Indica el número de línea en que encuentras la información:

- a. El término ingeniero en software es muy limitado en cuanto a su extensión. Lin N°.....
- b. El desorallador de programas es un especialista en determinadas áreas del desarrollo. Lin N°.....
- c. Quién escribe el texto esta confundido con lo que comprende el concepto de ingeniero en computación. Lin N°
- d. La ingeniería electrónica utiliza componentes electrónicos lineales y no activos. Lin N°.....
- e. Los ingenieros electrónicos tienen tareas de programación en lenguajes de de bajo nivel. Lin N°.....

b. Traduzcalas.

4. Anota en castellano el referente/s de las palabras subrayadas y resaltadas en el texto: 1,5pto.

- a. ... who...
- b... this way ...
- c. ...it...
- d. which ...
- e. ... them ...

5. Subraya el sustantivo nucleo y traduzca las siguientes frases nominales

- a. *a software engineer*
- b. *a more general term*
- c. *the software development process*
- d. *the Software Development Life Cycle*
- e. *the word computing*

6. Subraya el verbo de cada oración. Indica el tiempo verbal (presente, pasado o futuro), singular o plural. Traduzcalas.

- a. Recently I wrote a post about moving from Electronics to Software Engineering you can read here.
- b. *Systems engineering deals with work-processes, optimization methods, and risk management tools in such projects.*
- c. ...the other definitions we'll cover next.



- 7. Diga si la siguiente oracion expresa igualdad, comparacion o superioridad. Subraya el adjetivo que justifica tu respuesta. Traduzca.**

Companies often use the term to make emphasis on the analytical skills rather than implementation skills.